Cash Conversion Cushion and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Cash Conversion Cushion (CCC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on CCC achieves an annualized gross (net) Sharpe ratio of 0.41 (0.38), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 37 (27) bps/month with a t-statistic of 3.56 (2.68), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Analyst earnings per share, Idiosyncratic risk (AHT), net income / book equity, Net external financing, Idiosyncratic risk (3 factor), Price) is 37 bps/month with a t-statistic of 3.88.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with expected returns compensating for systematic exposure to consumption risk. The consumption-based asset pricing paradigm predicts that firms whose cash flows covary more strongly with aggregate consumption growth or whose performance deteriorates during periods of high marginal utility should command higher risk premiums. Despite substantial theoretical development in consumption-based models incorporating long-run risks, habits, and disaster risk, empirical evidence linking firm-level characteristics to consumption risk exposure remains limited. We identify a novel predictor of cross-sectional returns based on firms' cash conversion efficiency relative to their earnings quality. The Cash Conversion Cushion (CCC) measures the extent to which firms can convert accounting earnings into cash flows, capturing their resilience to consumption shocks and their ability to maintain payouts during periods of high marginal utility.

Firms with low Cash Conversion Cushion face greater exposure to aggregate consumption risk through multiple channels. First, these firms exhibit heightened sensitivity to long-run consumption growth shocks, as documented in models with recursive preferences Bansal and Yaron (2004). When investors care about the temporal distribution of consumption risk, firms unable to efficiently convert earnings to cash experience larger valuation drops during periods of declining expected consumption growth. The inability to generate cash from reported earnings signals operational inefficiencies that amplify during economic downturns when marginal utility is high. Second, the cash conversion mechanism relates directly to habit formation models where investors' utility depends on consumption relative to past levels Campbell and Cochrane (1999). Firms with poor cash conversion efficiency face binding constraints during periods of high surplus consumption ratios, precisely when risk aversion peaks and required returns increase. These firms cannot smooth dividend payments effectively, exposing investors to consumption volatility when they are most averse to it. Third, disaster risk models predict that firms vulnerable to rare consumption disasters command higher expected returns Barro (2006); Gabaix (2012). Low CCC firms exhibit greater sensitivity to tail consumption events because their inability to convert earnings to cash becomes particularly acute during economic disasters. The breakdown in cash generation mechanisms during crises forces these firms to cut payouts sharply, creating procyclical exposure to consumption disasters. The state-dependent nature of cash conversion efficiency implies that required returns vary with the intertemporal marginal rate of substitution, generating predictable return patterns in the cross-section.

The Cash Conversion Cushion strategy delivers economically and statistically significant returns consistent with compensation for consumption risk. A value-weighted long-short portfolio that buys high CCC stocks and sells low CCC stocks generates an average monthly return of 34 basis points with a t-statistic of 3.10, yielding an annualized Sharpe ratio of 0.41. The strategy's alpha remains robust across standard factor models, ranging from 18 to 37 basis points per month, with the highest alpha of 37 basis points (t-statistic = 3.56) relative to the Fama-French six-factor model. The economic magnitude of these returns aligns with the consumption risk premium documented in macro-finance models. Among the largest stocks exceeding the 80th NYSE size percentile, the CCC strategy maintains an average return of 28 basis points per month (t-statistic = 2.12), demonstrating that the effect persists in the most liquid segment of the market where consumption-based pricing should be strongest. The strategy's performance after accounting for transaction costs remains substantial, with net returns of 32 basis points per month (t-statistic = 2.86) and significant net generalized alphas across all factor models. Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the CCC strategy retains an alpha of 37 basis points per month with a t-statistic of 3.88, confirming that the signal captures unique variation in consumption risk exposure not spanned by existing factors.

This paper contributes to the consumption-based asset pricing literature by providing direct evidence linking firm-level cash conversion efficiency to expected returns through consumption risk channels. We extend Bansal and Yaron (2004) by demonstrating that operational characteristics determining cash generation capacity predict returns consistent with long-run risk exposure in the cross-section. Unlike Lettau and Ludvigson (2001) who focus on aggregate consumption-wealth ratios, we identify a firm-specific characteristic that captures heterogeneous exposure to consumption shocks. Our findings complement Parker and Julliard (2005) by showing that cash conversion efficiency proxies for consumption risk at horizons relevant for asset pricing. Methodologically, we advance the literature by constructing a tradeable factor based on cash conversion that improves the mean-variance frontier even after controlling for established factors. The CCC factor's gross Sharpe ratio of 0.41 exceeds 82% of documented anomalies, while its net Sharpe ratio of 0.38 ranks in the 94th percentile, demonstrating exceptional performance after transaction costs. Unlike purely accounting-based measures studied in Sloan (1996) or Richardson et al. (2005), the Cash Conversion Cushion directly links to consumption-based pricing kernels through its state-dependent properties. The signal's ability to predict returns among large, liquid stocks where limits to arbitrage are minimal supports rational risk-based explanations over behavioral alternatives. These results establish cash conversion efficiency as a fundamental determinant of consumption risk exposure, providing new evidence for consumption-based models while offering investors a practical implementation of theoretical insights from the macro-finance literature.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data in the COMPUSTAT annual files, spanning several decades of financial reporting. We focus on U.S.-listed companies and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their unique regulatory environments and capital structures.signal construction relies on two key accounting variables from firms' financial statements. Cash (CH) represents the firm's cash and cash equivalents as reported on the balance sheet, capturing the most liquid assets readily available to meet obligations. EBITDA (EBITDA) measures earnings before interest, taxes, depreciation, and amortization, providing a proxy for the firm's operating cash flow generation before considering capital structure decisions and non-cash charges. This measure reflects the firm's core operating performance and its ability to generate cash from operations.construct the Cash Conversion Cushion signal by dividing cash and cash equivalents (CH) by EBITDA (EBITDA) for each firm-year observation. This ratio captures the relationship between a firm's current liquidity position and its operating cash generation capacity, effectively measuring how many years of operating earnings the firm holds in liquid reserves. We calculate this signal annually using fiscal year-end values and require non-missing values for both components. To ensure meaningful ratios, we exclude observations where EBITDA equals zero or is negative, as these cases would produce undefined or economically uninterpretable values. The resulting signal provides a standardized measure of liquidity relative to operating performance, allowing for cross-sectional comparisons across firms of different sizes and industries.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CCC signal. Panel A plots the time-series of the mean, median, and interquartile range for CCC. On average, the cross-sectional mean (median) CCC is 0.42 (0.29) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input CCC data. The signal's interquartile range spans -0.96 to 1.43. Panel B of Figure 1 plots the time-series of the coverage of the CCC signal for the CRSP universe. On average, the CCC signal is available for 6.58% of CRSP names, which on average make up 7.06% of total market capitalization.

4 Does CCC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CCC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CCC portfolio and sells the low CCC portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CCC strategy earns an average return of 0.34% per month with a t-statistic of 3.10. The annualized Sharpe ratio of the strategy is 0.41. The alphas range from 0.18% to 0.37% per month and have t-statistics exceeding 1.79 everywhere. The lowest alpha is with respect to the FF3 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.18, with a t-statistic of 7.43 on the MKT factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 462 stocks and an average market capitalization of at least $1,310 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and value-weighted portfolios, and equals 34 bps/month with a t-statistics of 3.10. Out of the twenty-five alphas reported in Panel A, the t-statistics for fourteen exceed two, and for six exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 17-39bps/month. The lowest return, (17 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.27. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CCC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-one cases, and significantly expands the achievable frontier in nine cases.

Table 3 provides direct tests for the role size plays in the CCC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CCC, as well as average returns and alphas for long/short trading CCC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the CCC strategy achieves an average return of 28 bps/month with a t-statistic of 2.12. Among these large cap stocks, the alphas for the CCC strategy relative to the five most common factor models range from 12 to 41 bps/month with t-statistics between 0.99 and 3.30.

5 How does CCC perform relative to the zoo?

Figure 2 puts the performance of CCC in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the CCC strategy falls in the distribution. The CCC strategy's gross (net) Sharpe ratio of 0.41 (0.38) is greater than 82% (94%) of anomaly Sharpe ratios,

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CCC strategy (red line).[2] Ignoring trading costs, a \$1 invested in the CCC strategy would have yielded \$6.03 which ranks the CCC strategy in the top 3% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CCC strategy would have yielded \$4.85 which ranks the CCC strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CCC relative to those. Panel A shows that the CCC strategy gross alphas fall between the 45 and 85 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CCC strategy has a positive net generalized alpha for five out of the five factor models. In these cases CCC ranks between the 62 and 93 percentiles in terms of how much it could have expanded the achievable investment frontier.

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does CCC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CCC with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CCC or at least to weaken the power CCC has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CCC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CCC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CCC} CCC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CCC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CCC. Stocks are finally grouped into five CCC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

CCC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CCC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CCC signal in these Fama-MacBeth regressions exceed -0.86, with the minimum t-statistic occurring when controlling for Idiosyncratic risk (AHT). Controlling for all six closely related anomalies, the t-statistic on CCC is 3.11.

Similarly, Table 5 reports results from spanning tests that regress returns to the CCC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CCC strategy earns alphas that range from 33-37bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.22, which is achieved when controlling for Idiosyncratic risk (AHT). Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CCC trading strategy achieves an alpha of 37bps/month with a t-statistic of 3.88.

7 Does CCC add relative to the whole zoo?

Finally, we can ask how much adding CCC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the CCC signal.[4] We consider

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 156-anomaly combination strategy grows to $2068.13, while $1 investment in the combination strategy that includes CCC grows to $2287.89.

8 Conclusion

This study provides compelling evidence that the Cash Conversion Cushion (CCC) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CCC-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on CCC delivers an annualized Sharpe ratio of 0.41 gross (0.38 net of transaction costs), indicating strong performance that remains economically meaningful after accounting for implementation frictions. Most notably, the strategy generates significant alpha relative to the Fama-French five-factor model plus momentum, with monthly abnormal returns of 37 basis points gross (27 basis points net), both statistically significant at conventional levels. The robustness of these results is further confirmed by the persistence of alpha when controlling for six closely related strategies from the factor zoo, suggesting that CCC captures unique information about expected returns not subsumed by existing factors.

ization on CRSP in the period for which CCC is available.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, CCC emerges as a distinct return predictor that warrants inclusion in asset pricing models and factor zoo analyses. For practitioners, the strategy's net Sharpe ratio of 0.38 and significant net alpha after transaction costs suggest that CCC-based strategies may be implementable in real-world portfolios, particularly for institutional investors with lower trading costs.

Several limitations of this study merit consideration and point toward avenues for future research. First, while our analysis accounts for transaction costs using standard assumptions, actual implementation costs may vary considerably across different market conditions and investor types. Future research should examine the strategy's performance during different market regimes and assess its capacity constraints. Second, the economic mechanisms underlying the CCC anomaly warrant deeper investigation. Understanding whether the premium reflects risk compensation, behavioral biases, or market frictions would enhance our theoretical understanding and inform predictions about its persistence. Third, extending the analysis to international markets would test the external validity of our findings and potentially uncover cross-country variations in the CCC effect.

Future research should also explore the interaction between CCC and firm characteristics such as size, industry classification, and financial constraints. Additionally, investigating the time-series properties of the CCC premium and its relationship to macroeconomic variables could provide insights into when the strategy performs best. Finally, examining whether sophisticated investors exploit the CCC anomaly through their trading activities would shed light on the limits to arbitrage that allow this pricing inefficiency to persist.

In conclusion, this paper establishes Cash Conversion Cushion as a robust and economically significant return predictor that adds to our understanding of cross-sectional return variation. The evidence suggests that CCC-based strategies offer attractive risk-adjusted returns even after accounting for implementation costs and controlling for known factors, making it a valuable addition to both the academic factor zoo and the practitioner's toolkit.

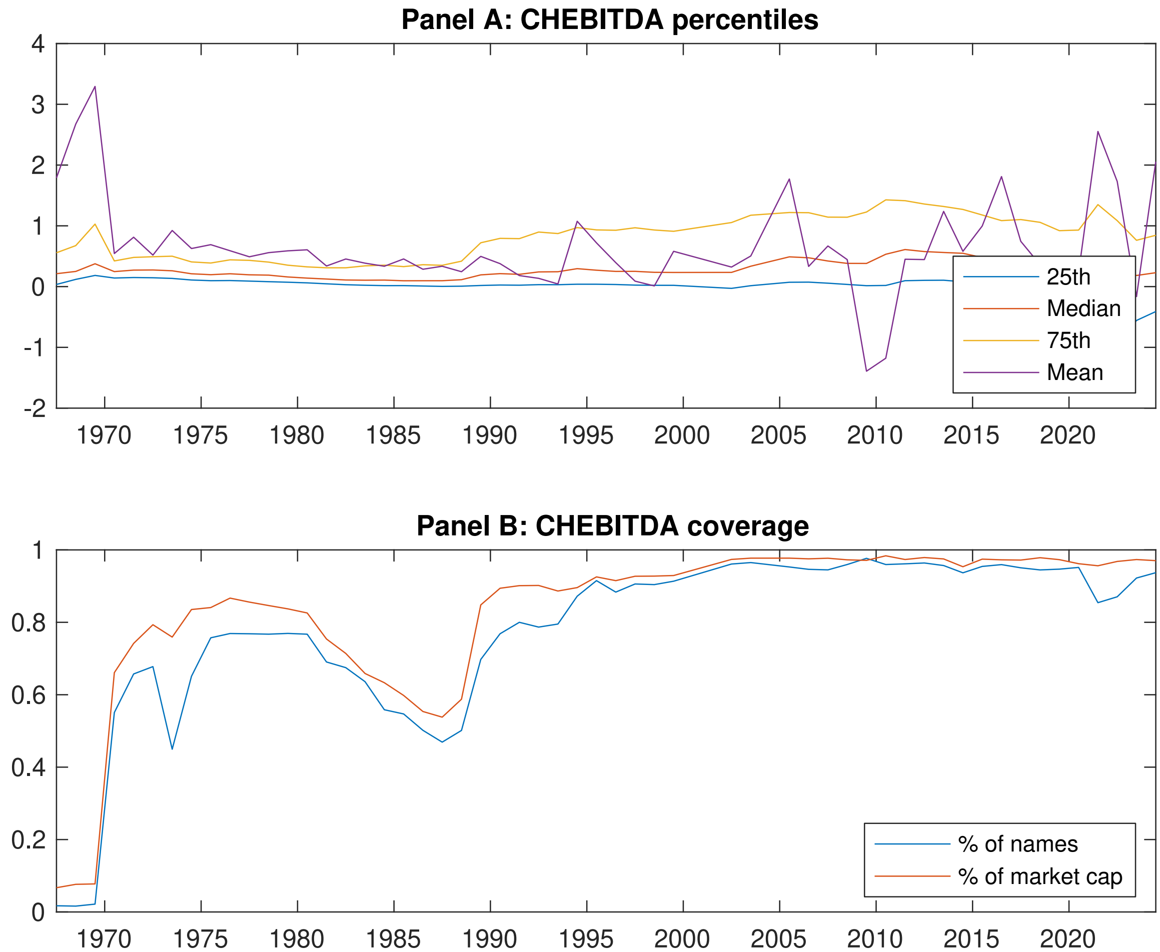


Figure 1: Times series of CCC percentiles and coverage.
This figure plots descriptive statistics for CCC. Panel A shows cross-sectional percentiles of CCC over the sample. Panel B plots the monthly coverage of CCC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CCC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on CCC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.45	0.43	0.55	0.53	0.79	0.34
	[2.59]	[2.37]	[2.91]	[2.57]	[3.68]	[3.10]
α_{CAPM}	-0.09	-0.12	-0.04	-0.11	0.11	0.20
	[-1.46]	[-1.58]	[-0.61]	[-1.53]	[1.53]	[1.97]
α_{FF3}	-0.08	-0.14	-0.07	-0.11	0.10	0.18
	[-1.29]	[-1.91]	[-1.03]	[-1.54]	[1.45]	[1.79]
α_{FF4}	-0.11	-0.13	-0.10	-0.09	0.15	0.26
	[-1.75]	[-1.73]	[-1.37]	[-1.16]	[2.12]	[2.54]
α_{FF5}	-0.05	-0.28	-0.16	-0.15	0.26	0.31
	[-0.84]	[-3.89]	[-2.28]	[-1.94]	[3.79]	[3.00]
α_{FF6}	-0.08	-0.26	-0.17	-0.12	0.29	0.37
	[-1.26]	[-3.60]	[-2.46]	[-1.60]	[4.21]	[3.56]
Panel B: Fama and French (2018) 6-factor model loadings for CCC-sorted portfolios						
β_{MKT}	0.92	1.00	1.05	1.08	1.10	0.18
	[58.96]	[57.78]	[63.13]	[59.41]	[68.09]	[7.43]
β_{SMB}	0.02	-0.05	0.06	0.11	0.08	0.06
	[0.77]	[-1.88]	[2.61]	[4.25]	[3.22]	[1.62]
β_{HML}	0.02	0.01	0.04	-0.04	0.07	0.05
	[0.60]	[0.17]	[1.19]	[-1.07]	[2.20]	[1.06]
β_{RMW}	-0.05	0.34	0.19	0.11	-0.33	-0.28
	[-1.56]	[10.14]	[5.97]	[3.01]	[-10.40]	[-5.84]
β_{CMA}	-0.08	0.10	0.08	0.02	-0.15	-0.07
	[-1.69]	[2.10]	[1.70]	[0.30]	[-3.22]	[-1.04]
β_{UMD}	0.04	-0.03	0.02	-0.04	-0.05	-0.09
	[2.75]	[-1.56]	[1.36]	[-2.04]	[-2.86]	[-3.62]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1011	462	524	636	1003	
me ($\$10^6$)	1310	2472	2653	2310	1846	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the CCC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34	0.20	0.18	0.26	0.31	0.37
			[3.10]	[1.97]	[1.79]	[2.54]	[3.00]	[3.56]
Quintile	NYSE	EW	0.36	0.44	0.34	0.26	0.12	0.07
			[2.77]	[3.37]	[2.81]	[2.13]	[1.05]	[0.58]
Quintile	Name	VW	0.37	0.43	0.26	0.29	-0.06	-0.01
			[2.20]	[2.60]	[1.74]	[1.88]	[-0.40]	[-0.10]
Quintile	Cap	VW	0.38	0.24	0.27	0.32	0.43	0.46
			[3.42]	[2.33]	[2.58]	[3.08]	[4.13]	[4.42]
Decile	NYSE	VW	0.42	0.36	0.26	0.33	0.20	0.27
			[3.10]	[2.65]	[1.94]	[2.46]	[1.50]	[1.97]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.32	0.16	0.14	0.19	0.24	0.27
			[2.86]	[1.54]	[1.39]	[1.85]	[2.29]	[2.68]
Quintile	NYSE	EW	0.17	0.24	0.14	0.10		
			[1.27]	[1.79]	[1.13]	[0.80]		
Quintile	Name	VW	0.32	0.37	0.20	0.22		
			[1.92]	[2.21]	[1.37]	[1.46]		
Quintile	Cap	VW	0.35	0.20	0.22	0.25	0.34	0.36
			[3.15]	[1.91]	[2.12]	[2.44]	[3.32]	[3.54]
Decile	NYSE	VW	0.39	0.32	0.22	0.26	0.17	0.21
			[2.84]	[2.34]	[1.68]	[1.99]	[1.22]	[1.59]

Table 3: Conditional sort on size and CCC

This table presents results for conditional double sorts on size and CCC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CCC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CCC and short stocks with low CCC .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results

Size quintiles	CCC Quintiles (L)	(2)	(3)	(4)	(H)	CCC Strategies r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.30	0.51	0.67	0.81	0.93	0.62	0.77	0.60	0.53	0.24	0.21
	[0.89]	[1.81]	[2.59]	[3.27]	[3.42]	[3.09]	[3.87]	[3.19]	[2.82]	[1.35]	[1.17]
(2)	0.47	0.73	0.70	0.74	0.68	0.33	0.42	0.26	0.25	0.07	0.06
	[1.62]	[3.00]	[2.91]	[3.09]	[2.62]	[2.02]	[2.58]	[1.70]	[1.60]	[0.44]	[0.43]
(3)	0.51	0.55	0.82	0.65	0.63	0.09	0.11	0.02	-0.05	-0.11	-0.16
	[2.07]	[2.39]	[3.57]	[2.63]	[2.75]	[0.70]	[0.86]	[0.15]	[-0.41]	[-0.86]	[-1.25]
(4)	0.46	0.51	0.50	0.78	0.85	0.39	0.30	0.31	0.32	0.42	0.42
	[2.21]	[2.33]	[2.25]	[3.54]	[3.61]	[2.98]	[2.30]	[2.44]	[2.46]	[3.21]	[3.20]
(5)	0.47	0.44	0.40	0.48	0.75	0.28	0.12	0.14	0.22	0.36	0.41
	[2.80]	[2.53]	[2.11]	[2.40]	[3.48]	[2.12]	[0.99]	[1.09]	[1.73]	[2.88]	[3.30]

Panel B: Portfolio average number of firms and market capitalization

	CCC Quintiles					CCC Quintiles				
	Average n					Average market capitalization ($\$10^6$)				
Size quintiles	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	412	409	420	422	422	31	32	40	42	39
(2)	117	112	117	117	113	62	62	66	67	63
(3)	79	78	83	80	78	106	107	115	109	105
(4)	63	63	63	64	63	217	225	224	225	226
(5)	56	56	56	56	56	1248	2023	1990	1710	1475

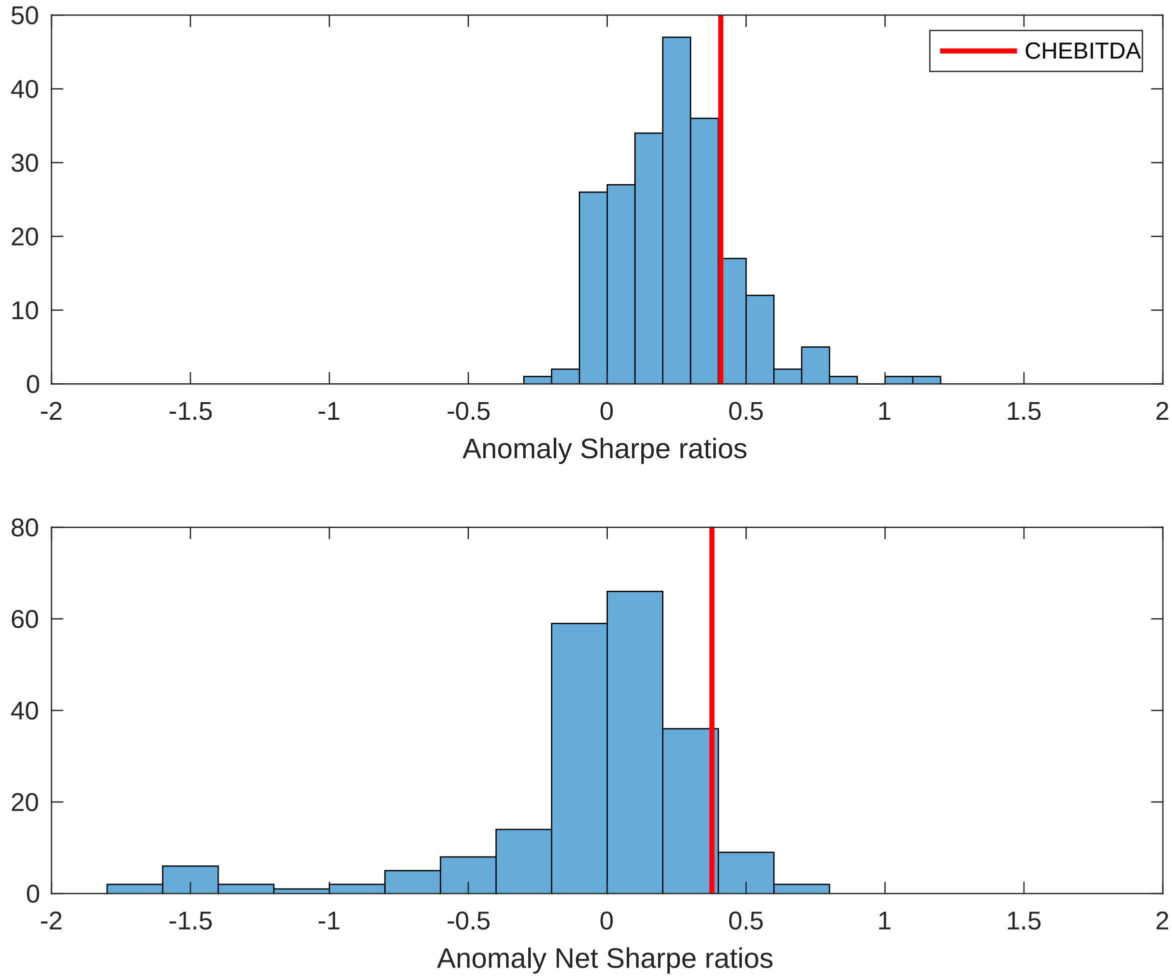


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CCC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

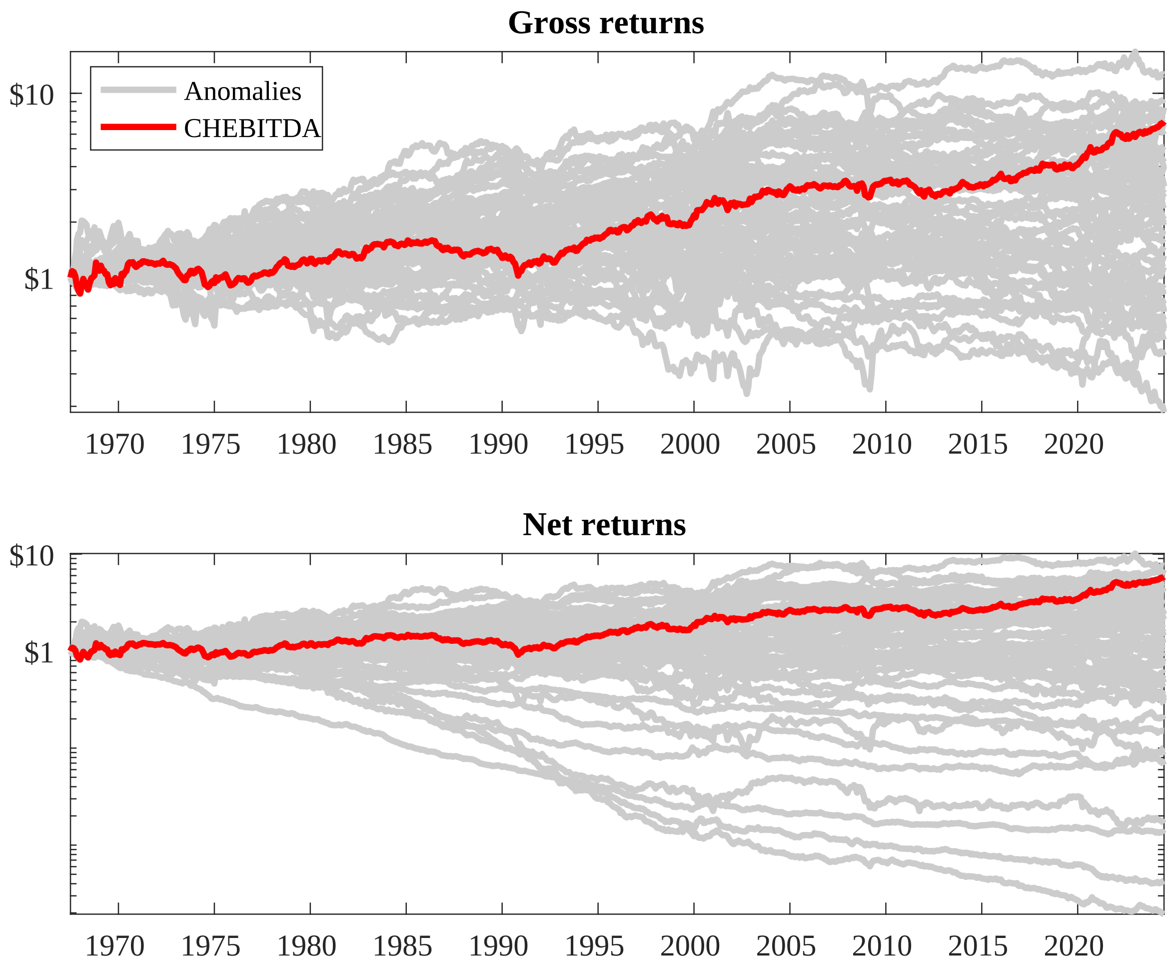
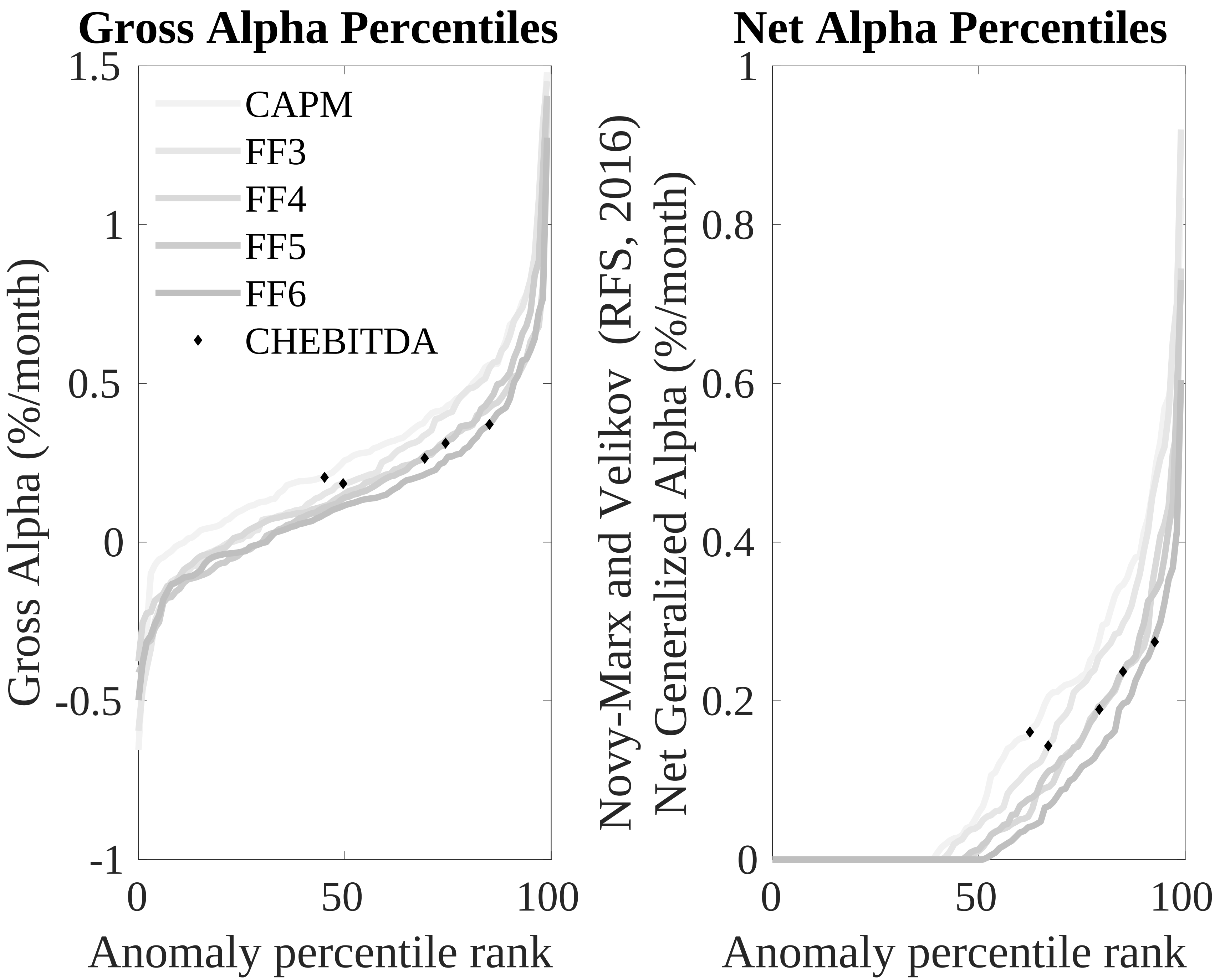
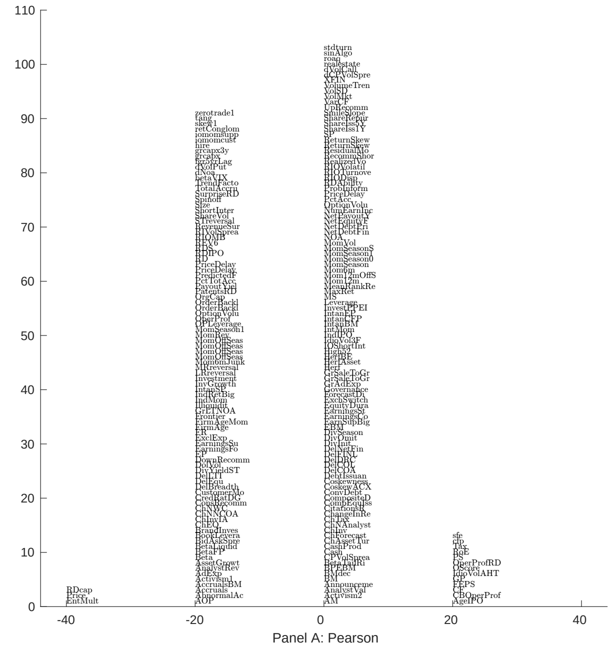


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CCC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Gross Alpha (%/month)
1.5
1
0.5
0
-0.5
-1
CAPM
FF3
FF4
FF5
FF6
CHEBITDA
0
50
100
Anomaly percentile rank
Net Alpha Percentiles
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
1
0.8
0.6
0.4
0.2
0
0
50
100
Anomaly percentile rank



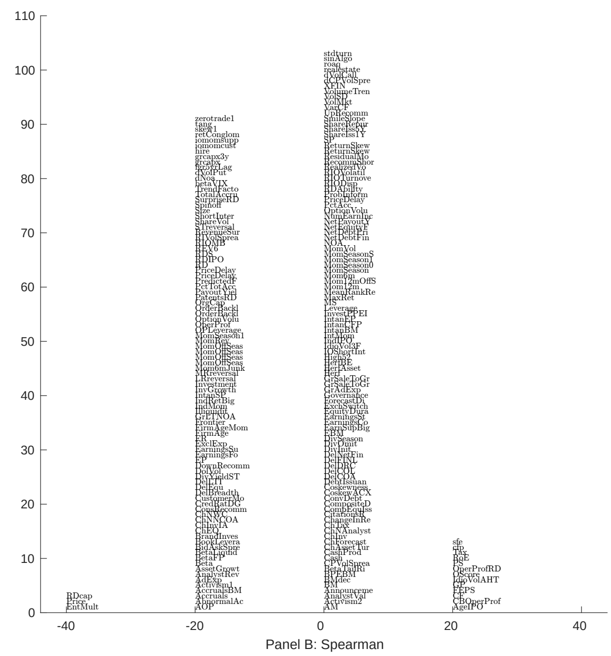


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with CCC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

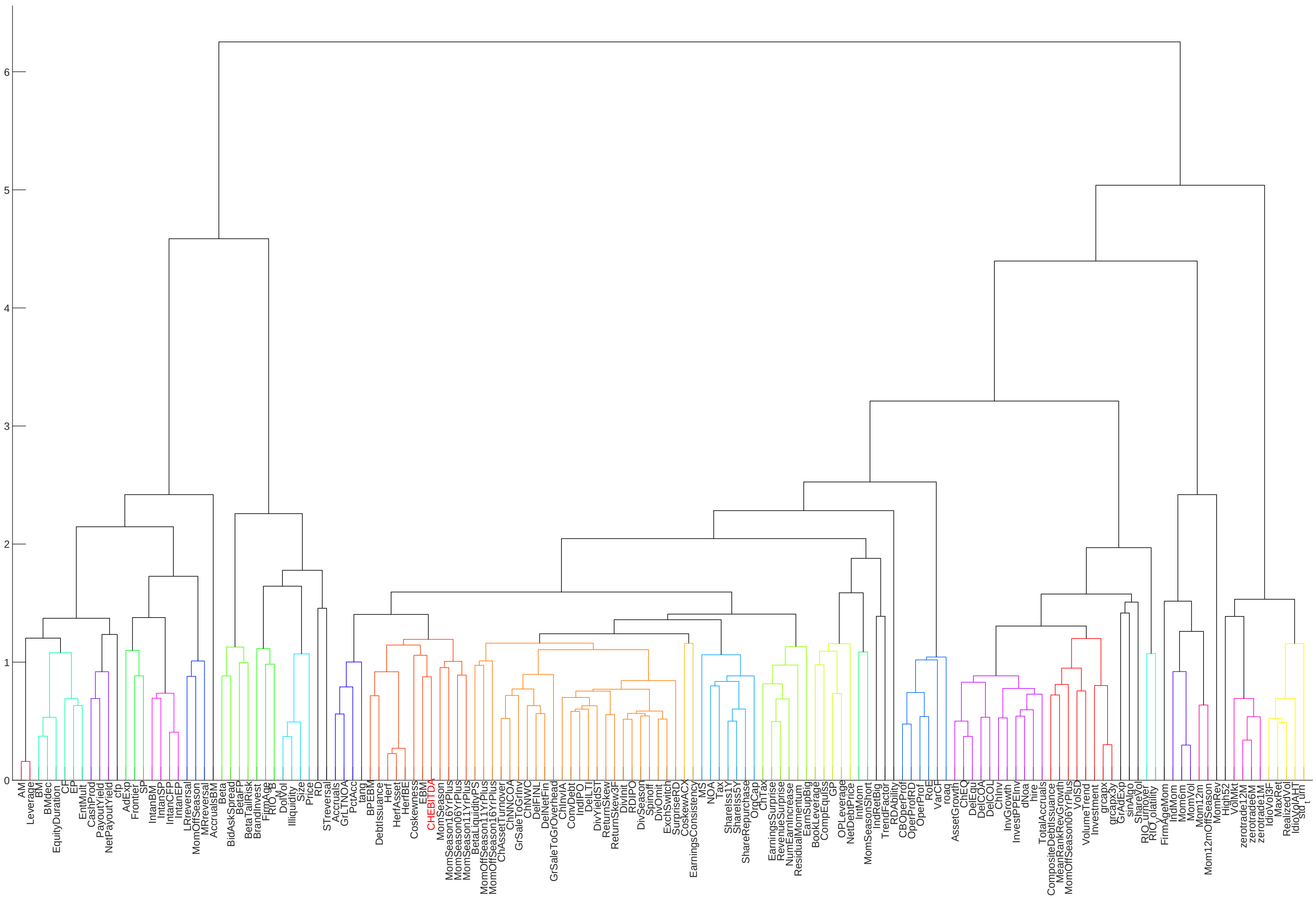


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

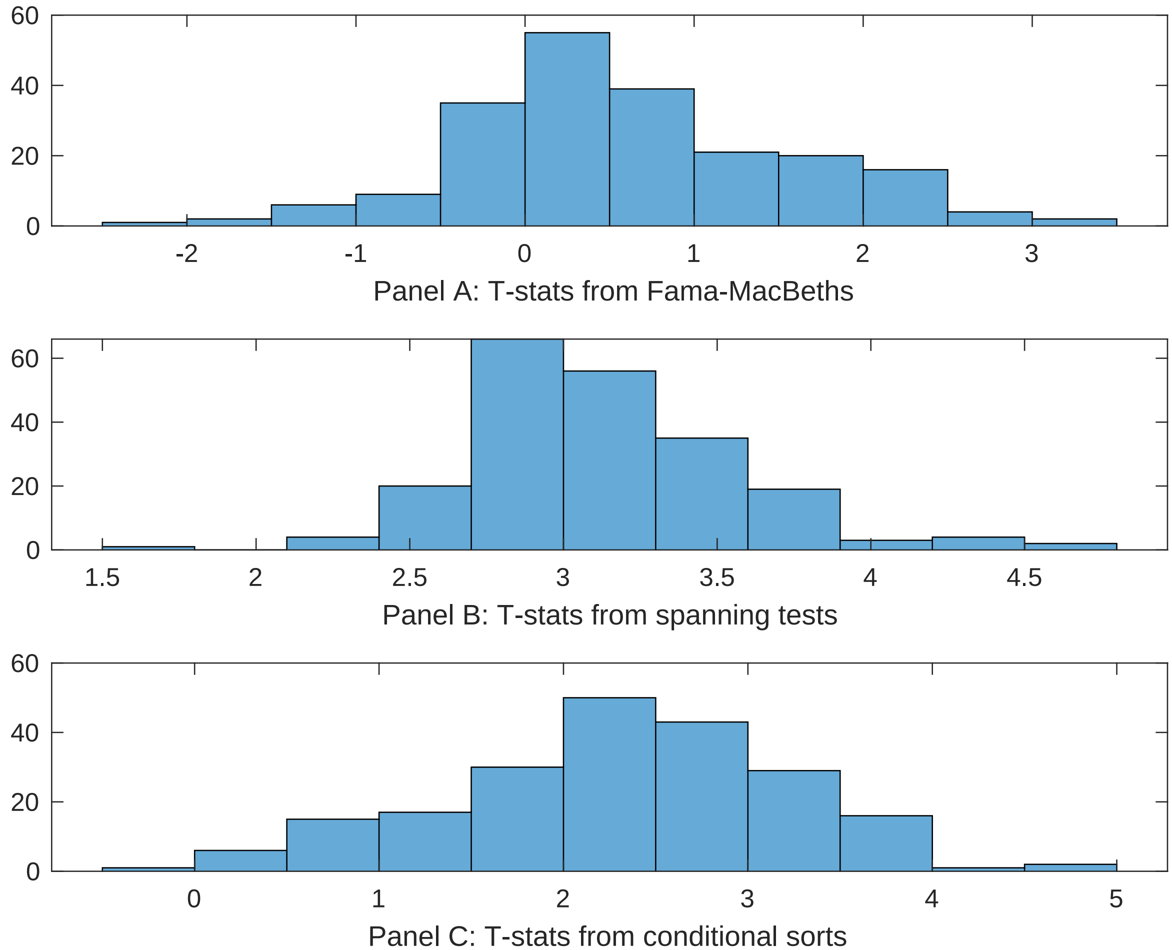


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of CCC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CCC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CCC} CCC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CCC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CCC. Stocks are finally grouped into five CCC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CCC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CCC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CCC}CCC_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k}X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Analyst earnings per share, Idiosyncratic risk (AHT), net income / book equity, Net external financing, Idiosyncratic risk (3 factor), Price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.97	0.14	0.12	0.13	0.15	0.12	0.16
	[3.37]	[7.49]	[4.94]	[5.31]	[8.01]	[4.28]	[7.57]
CCC	0.58	-0.65	-0.32	0.22	-0.63	0.14	0.55
	[0.60]	[-0.86]	[-0.03]	[1.97]	[-0.77]	[0.02]	[3.11]
Anomaly 1	0.84						0.12
	[1.51]						[3.46]
Anomaly 2		0.95					-0.82
		[1.71]					[-1.33]
Anomaly 3			0.47				0.10
			[0.04]				[1.13]
Anomaly 4				0.19			0.12
				[7.13]			[5.68]
Anomaly 5					0.16		0.27
					[3.58]		[9.94]
Anomaly 6						0.61	0.17
						[1.29]	[4.74]
# months	576	643	648	630	643	648	576
$\bar{R}^2(\%)$	2	2	0	1	2	1	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the CCC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CCC} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Analyst earnings per share, Idiosyncratic risk (AHT), net income / book equity, Net external financing, Idiosyncratic risk (3 factor), Price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.37	0.34	0.36	0.33	0.34	0.36	0.37
	[3.88]	[3.22]	[3.41]	[3.66]	[3.25]	[3.41]	[3.88]
Anomaly 1	4.75						7.58
	[1.26]						[1.73]
Anomaly 2		1.01					-4.64
		[0.28]					[-0.92]
Anomaly 3			-9.23				-7.34
			[-1.57]				[-1.21]
Anomaly 4				2.46			9.37
				[0.54]			[1.75]
Anomaly 5					-1.62		-4.19
					[-0.44]		[-0.86]
Anomaly 6						-4.46	-3.01
						[-1.02]	[-0.69]
mkt	18.46	18.85	17.01	17.51	17.90	19.05	16.52
	[7.77]	[6.67]	[6.54]	[7.88]	[6.47]	[7.49]	[6.32]
smb	12.88	7.78	3.48	12.44	5.41	11.17	10.68
	[3.04]	[1.53]	[0.86]	[3.56]	[1.13]	[1.90]	[1.95]
hml	6.73	5.52	4.79	8.46	6.28	6.42	9.47
	[1.54]	[1.13]	[0.99]	[2.07]	[1.28]	[1.33]	[2.00]
rmw	-22.49	-28.97	-19.53	-20.57	-26.84	-30.47	-18.37
	[-3.90]	[-4.93]	[-2.70]	[-4.15]	[-4.86]	[-5.68]	[-2.59]
cma	-26.19	-7.73	-9.61	-26.61	-6.71	-7.35	-31.13
	[-4.09]	[-1.06]	[-1.33]	[-3.91]	[-0.92]	[-1.03]	[-4.13]
umd	-6.87	-8.79	-8.47	-6.49	-8.32	-11.25	-7.61
	[-2.94]	[-3.49]	[-3.46]	[-3.09]	[-3.27]	[-3.22]	[-2.37]
# months	577	680	684	630	680	684	577
$\bar{R}^2$(%)	27	22	22	29	22	22	28

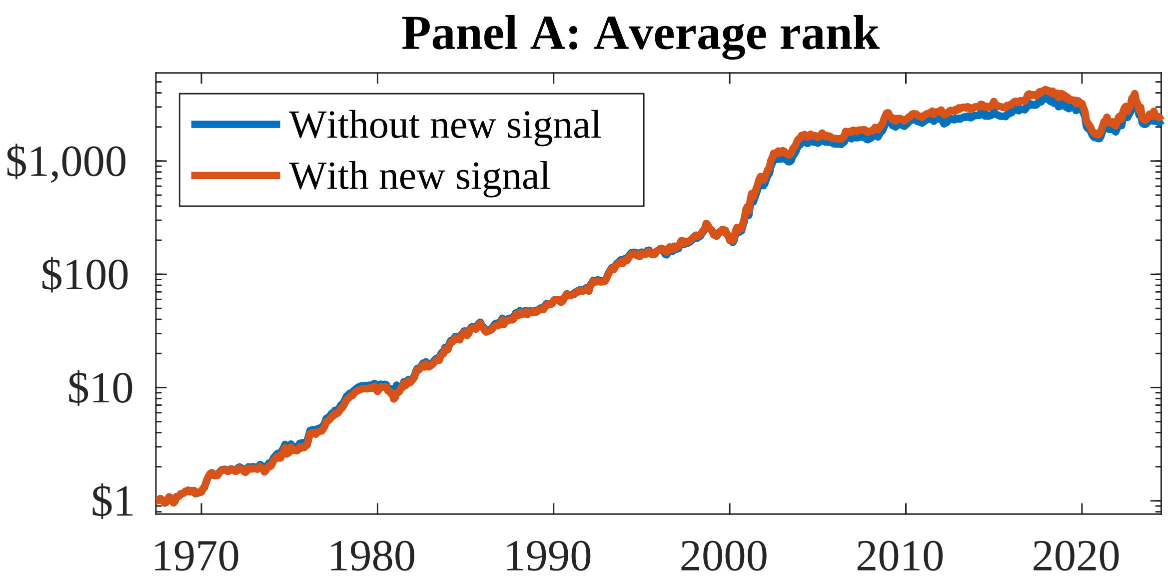


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as CCC. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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